



MASARAK

GROUP 2

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Abstract

This project presents a machine learning-based system designed to recommend the most suitable university majors for students based on their academic performance, personality traits, and social preferences. The model takes into account the student's overall GPA, core subject grades from high school, and answers to structured questions divided into academic, personal, and social categories. Using this input, the system predicts and ranks the top three university majors that best match the student's profile, providing results in descending order of compatibility. In addition to the recommendations, the system also explains the reasoning behind each suggested major, highlighting how the student's academic background and personal attributes align with the requirements and nature of the field. This approach aims to support students in making more informed and personalized decisions about their higher education pathways.

Methodology

The methodology of this project is divided into three main stages: data preparation, model training, and recommendation generation.

Data Preparation

Since no suitable dataset was available, we created a custom dataset from scratch. This step was one of the most challenging parts of the project. The dataset included academic data (overall GPA and core subject grades), as well as responses to structured questionnaires covering personal, social, and psychological aspects. Designing the dataset required simulating realistic distributions of grades and answers, ensuring diversity across student profiles, and balancing the classes (majors) to avoid bias. Despite the difficulties, this step was crucial for enabling the machine learning models to learn effectively.

Methodology

Feature Engineering

In the project, feature engineering was designed to transform raw student and program data into meaningful inputs for machine learning models. The process followed three main directions:

Academic Features

Extracted features from students' academic performance, such as GPA and core subject averages.

Created categorical bins (GPA ranges) and percentile scores to capture relative standing.

Computed ratios (core subject average) for additional insights.

Methodology

Interest-Based Features

Converted questionnaire responses into structured features.

For Likert-scale questions, normalized answers (1–5) into 0–1 values and mapped them to fields of interest using weighted contributions.

For multiple-choice questions, applied one-hot encoding to represent each selected option.

When available, used external weight mappings to translate selected options into contributions toward interest fields.

Normalized all interest scores into a 0–100 scale and identified the top interests for each student.

Eligibility Features

Determined program eligibility by combining branch requirements with minimum GPA thresholds.

Created binary features indicating whether a student is eligible for each program.

Calculated the total number of programs each student qualifies for.

Methodology

Model Training

After preprocessing the dataset (normalizing numerical values and encoding categorical responses), we experimented with several classification algorithms to evaluate their performance and gain insights into their suitability for our task:

- **Logistic Regression:** Implemented as a simple baseline model to observe how a linear approach performs on the dataset.
- **Gradient Boosting Classifier:** Tested for its potential to improve predictive accuracy through sequential tree building.
- **Random Forest:** Selected as the final model because of its robustness in handling both categorical and numerical features, its ability to reduce overfitting, and the consistently reliable accuracy it achieved during evaluation.

Although multiple models were tested, the **Random Forest** classifier was ultimately chosen as the final model, while the others were only explored experimentally for comparison.

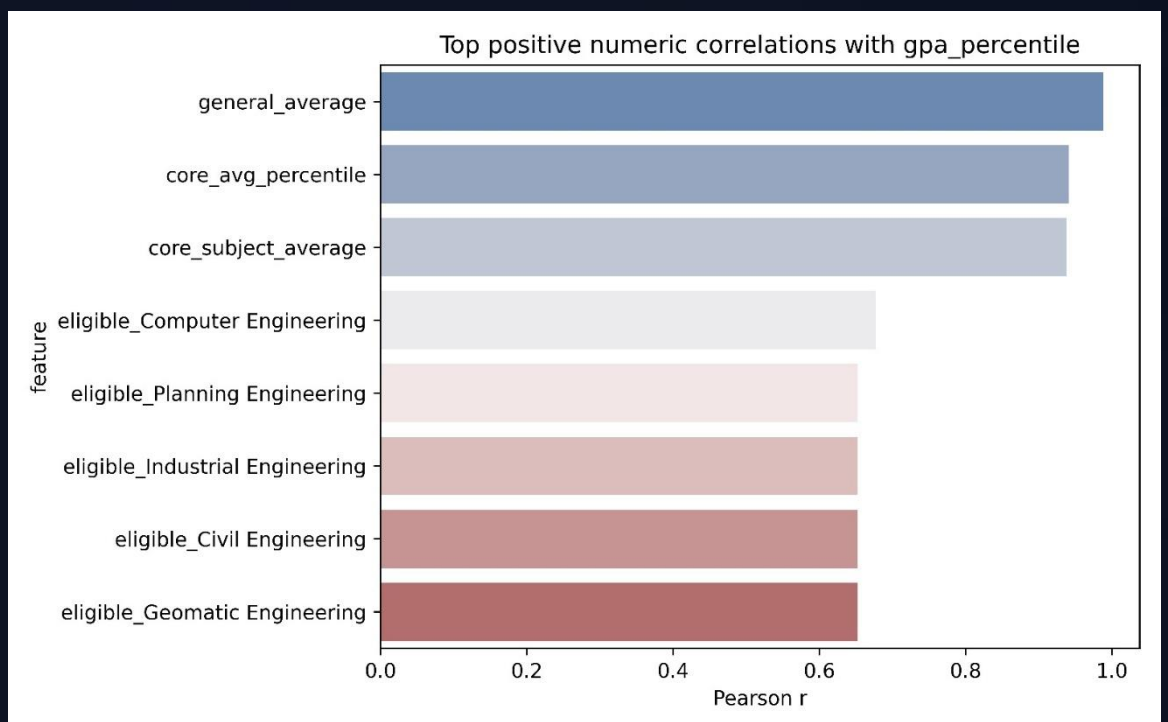
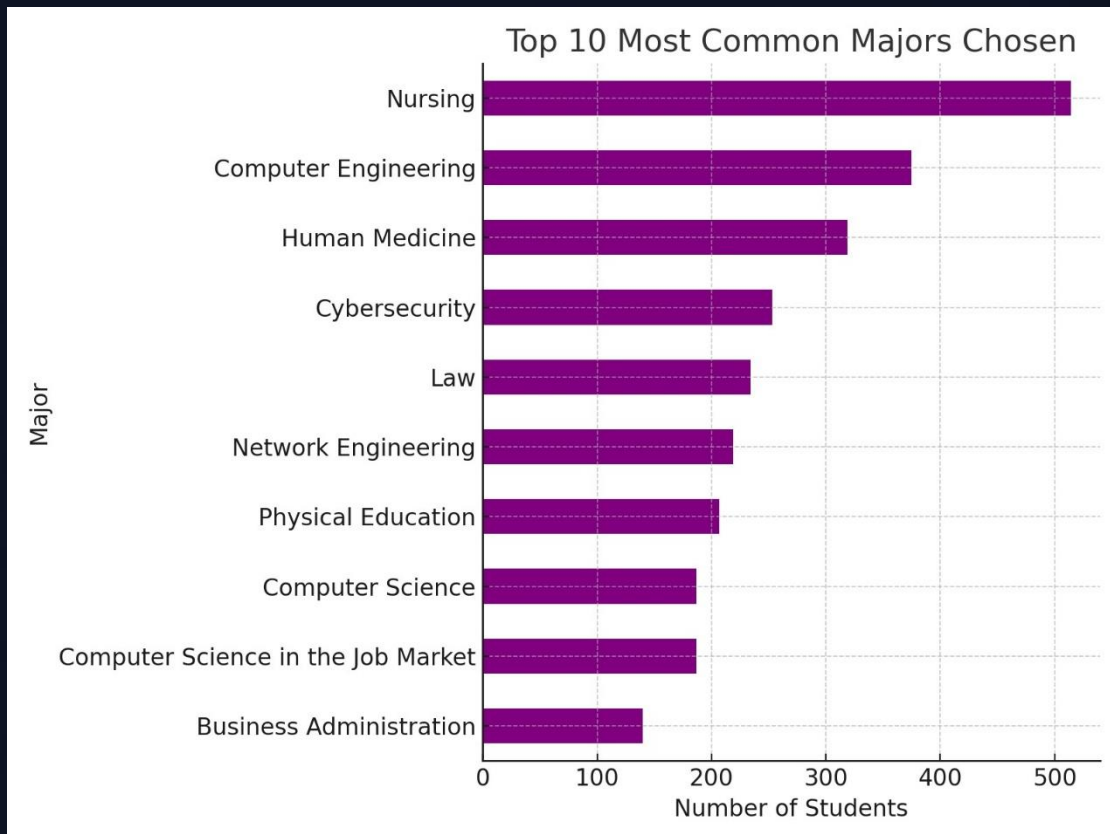
Methodology

Model Recommendation Generation

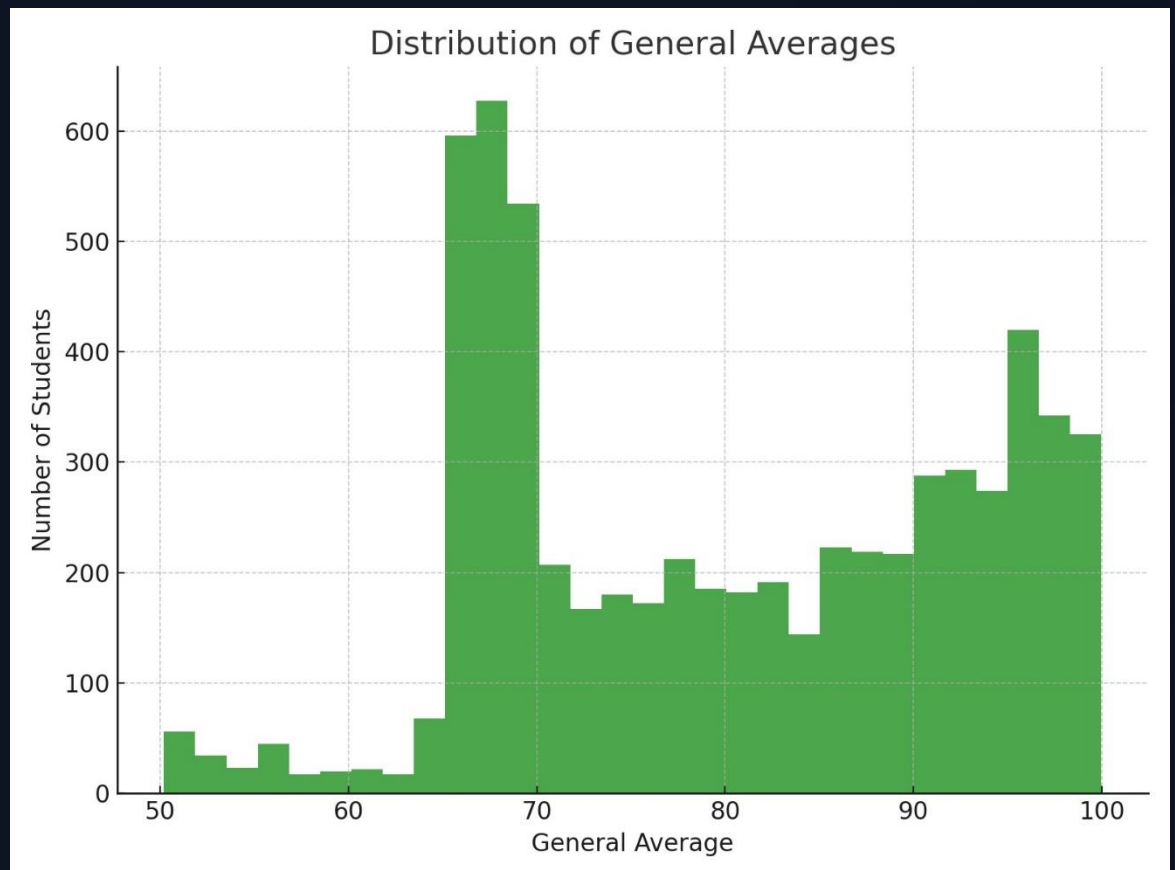
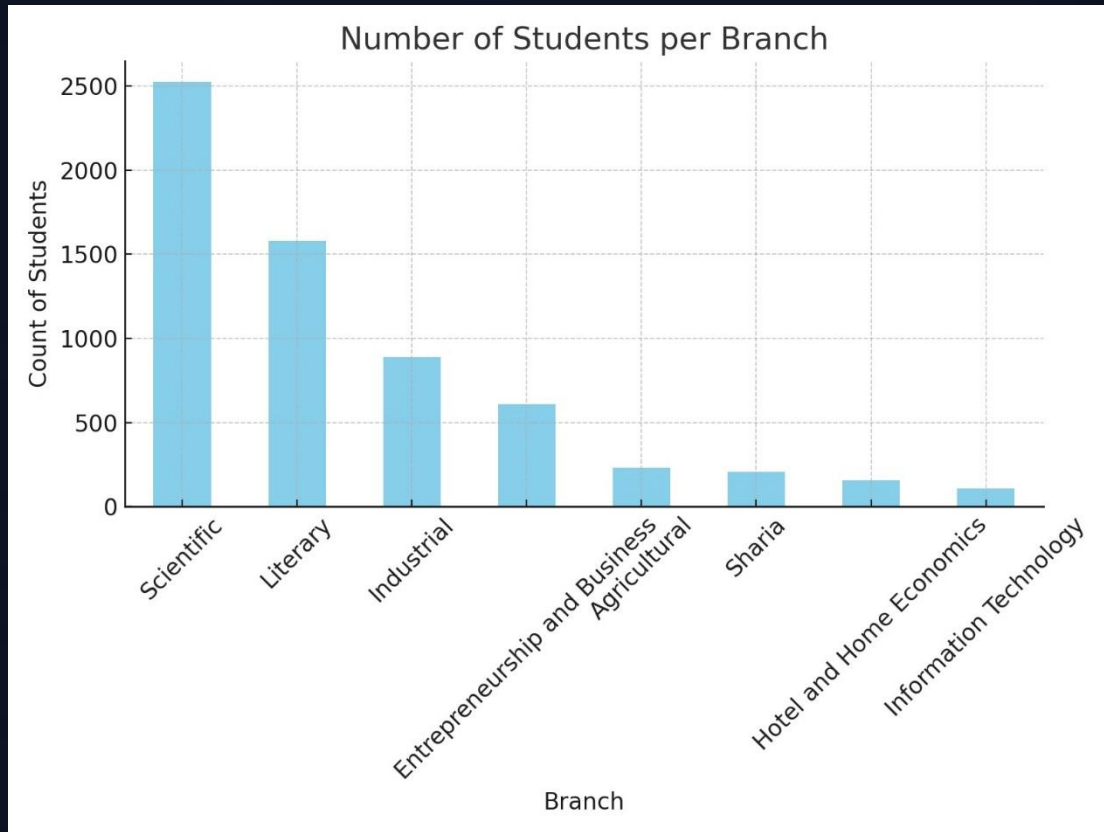
The selected model outputs probability scores for each possible university major. Based on these scores, the system ranks the top three majors from the most to the least suitable. To improve transparency, the model also provides explanations for why a specific major matches the student's profile, relying on feature importance derived from Gradient Boosting.

Through experimental comparison, the Gradient Boosting Classifier demonstrated the best overall performance in terms of accuracy, precision, and recall, and was therefore selected as the final model for the system.

Data Visualization

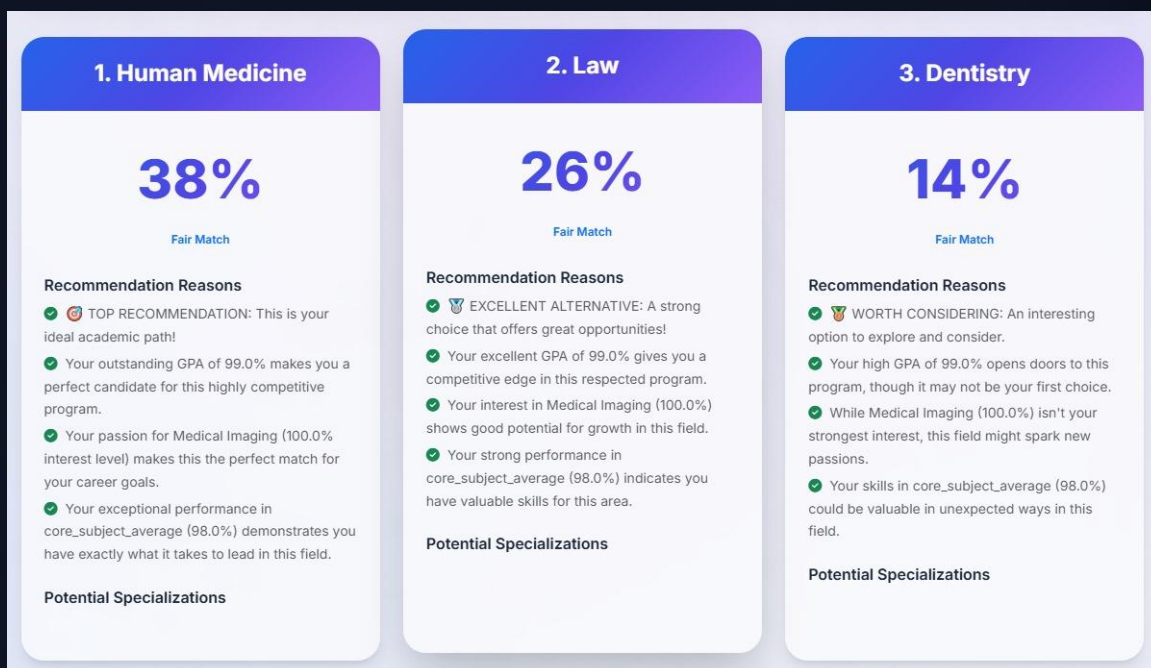


Data Visualization

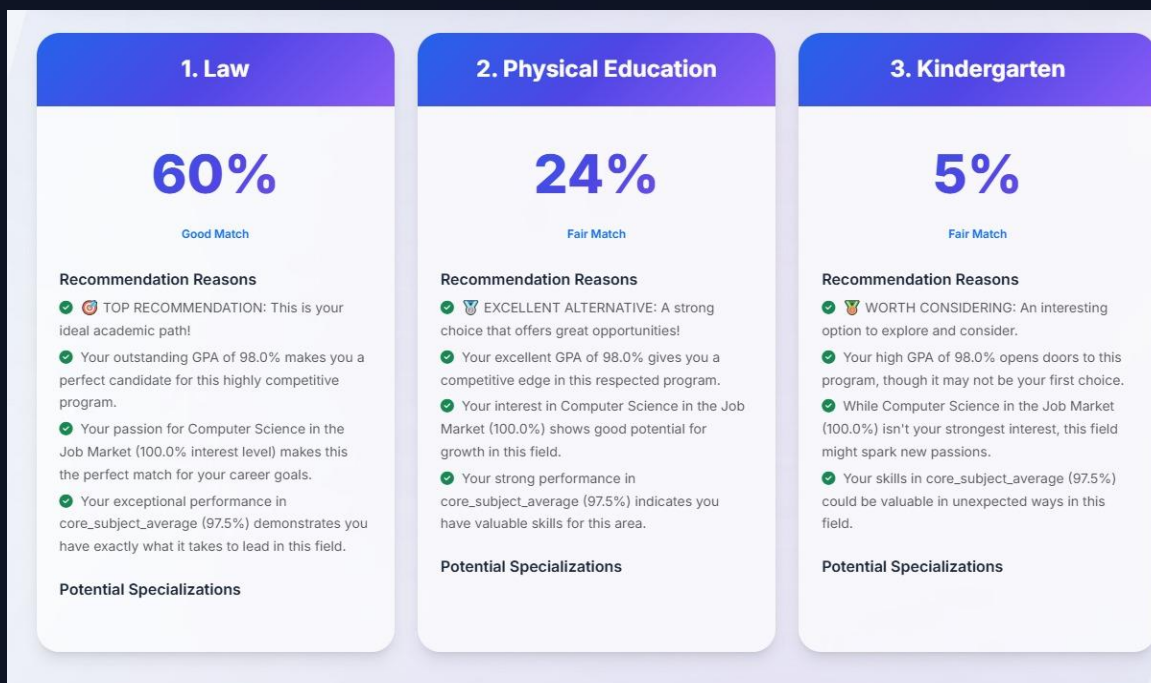


Result

Result of entering a scientific sample:



Result of entering a literary sample:



Result

As demonstrated in the previous slide, we provided two different samples across all criteria. The model then identified the top three most suitable specializations for each sample, ranked them according to relevance, and generated tailored recommendations accordingly.

Overall Results

During the development and evaluation of our University Recommendation System, we tested several output strategies to measure the accuracy and reliability of the model:

Single Major Recommendation

The model suggested only one major, which was deemed the most suitable for the student.

This approach achieved an accuracy of 73%.

Result

Top-3 Majors Recommendation (Adopted Approach)

The model suggested three majors, with the most appropriate major included among the three.

This option provided a balance between accuracy and usability, achieving 87% accuracy.

Based on these results, this approach was adopted as the final recommendation method.

Top-5 Majors Recommendation

The model suggested five majors, including the most appropriate one.

This approach achieved the highest accuracy of 95%, but with reduced precision in guiding students to a narrower set of choices.

In addition, we invited users to support the project's development by providing feedback on the major they are currently studying or the one they felt was best matched by the model's recommendations. This feedback loop plays a crucial role in enhancing the system's accuracy and ensuring continuous improvement.

Conclusions

The project successfully demonstrated that a machine learning model can recommend the top three most suitable university majors for students, ranked by likelihood and accompanied by explanations.

The interpretability aspect explaining why a given major is a good fit was essential in making the system practical and trustworthy for real use.

One of the most significant achievements was building the dataset from scratch. Since no suitable public dataset existed for this purpose, a synthetic (dummy) dataset had to be designed and constructed manually. This involved carefully defining majors, designing academic and personality/social features, and ensuring the dataset was realistic and balanced enough for training and evaluation.

Overall, the project highlighted both the potential and the limitations of AI-driven career guidance systems.

Challenges

a. Dataset Creation (Dummy Data)

The biggest challenge was the complete absence of a real dataset. All training and testing data had to be generated from scratch, which was time-consuming and complex.

Creating dummy data was not just about random numbers it required:

Defining the most common university majors.

Designing academic features (grades in core subjects) and personality/social questions that are realistically linked to those majors.

Simulating realistic correlations, e.g., strong Math/Physics for Engineering, strong Biology/People-oriented traits for Medicine.

Challenges

b. Data Reliability

Even with a carefully designed dummy dataset, the model cannot fully capture the diversity and unpredictability of real student populations.

In real deployment, questionnaire responses may be biased (students exaggerating or underestimating their traits).

c. Class Imbalance

Some majors naturally had fewer samples than others, which led to imbalanced performance across classes. Specialized majors (e.g., Civil Engineering, Nursing) showed lower F1 scores compared to more common ones.

Challenges

d. Model Explain ability

Feature importance plots provided insights, but students and counselors may still find it challenging to fully trust an AI-driven recommendation.

e. Generalization and Context

The model is tailored to one specific education system (local high school tracks). Applying it to other countries would require adapting the dataset and retraining the model.

f. Ethical Considerations

Since university major choice is a life-changing decision, the system should be presented as a supportive tool, not as a final authority. Students must retain autonomy, with AI serving as guidance.

Future Work

Collecting real data: The most critical next step is to gather real student data, including their academic records, questionnaire responses, chosen majors, and later academic performance. This would allow the model to be retrained and validated on actual outcomes, improving both accuracy and credibility.

Improved questionnaires: Refine the social/personal survey with validated psychometric tests to reduce subjectivity and bias.

Cross-context adaptation: Extend the model to other education systems by introducing modular data-preprocessing pipelines.

Ethical safeguards: Provide disclaimers and integrate the tool with human counselors, ensuring students receive balanced advice.

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